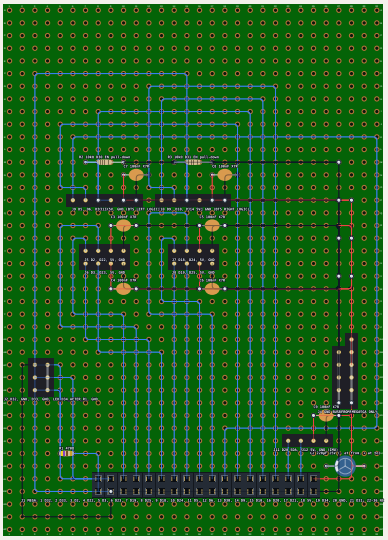
PERFBOARD DESIGNER · PROJECT SHEET

perfboard-corrected 5

Perfboard · 30 × 42 holes · 76.2 × 106.7 mm · November 1, 2026

● No problems found, 1 thing to look at: see Checks

25	19	22	66	3	114
parts	kinds	nets	wires	solder paths	joints

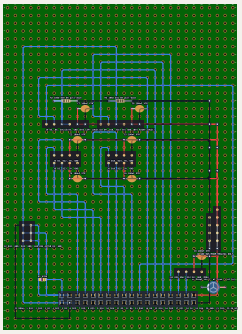


01 Board pictures 02 Components 03 Shopping list 04 Nets 05 Checks

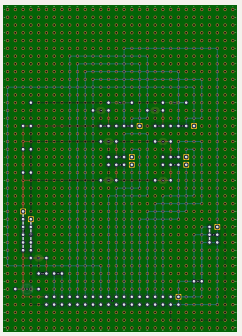
01 Board

As the editor draws it: the top with each part's ref, and the solder side as you'll hold it to solder, every joint marked.

Top · the component side



Solder side · seen from below



02 Components

25 parts on the board, by ref. Hole: where pin 1 goes.

REF	PART	VALUE	PINS	HOLE
C2	Electrolytic Capacitor Passives › Capacitors	220µF 25V (- at from, + at to)	2	AK26
C3	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	R9
C4	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	W9
C5	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	R16
C6	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	W16
C7	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	N10
C8	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	N17
C9	Ceramic Capacitor Passives › Capacitors	100nF X7R	2	AG25
J1	Female Header (2.54 mm) (2 × 18) Connectors	MEGA: 1 D32, 2 D33, 3 D2, 4 D22, 5 D3, 6 D23, 7 D19, 8 D25, 9 D18, 10 D24, 11 D5, 12 D6, 13 D30, 14 D9, 15 D10, 16 D20, 17 D21, 18 5V, 19 D34, 20 GND, 21 D31; 22-36 NC	36	AL8
J2	Header Pins (3 × 2) Connectors	D32, GND, D33, GND, LED_D34_AFTER_R1, GND	6	AC3
J3	Header Pins (6 × 1) Connectors	5V BUS FROM MEGA ONLY	6	AA28
J4	Header Pins (5 × 1) Connectors	GND BUS FROM MEGA	5	AB27
J5	Header Pins (1 × 4) Connectors	D2, D22, 5V, GND	4	T7
J6	Header Pins (1 × 4) Connectors	D3, D23, 5V, GND	4	U7

	REF	PART	VALUE	PINS	HOLE
	J7	Header Pins (1 × 4) Connectors	D18, D24, 5V, GND	4	T14
	J8	Header Pins (1 × 4) Connectors	D19, D25, 5V, GND	4	U14
	J9	Header Pins (1 × 4) Connectors	D5, D6, D30, D30	4	P6
	J10	Header Pins (1 × 4) Connectors	D9, D10, D31, D31	4	P13
	J11	Header Pins (1 × 2) Connectors	D20 SDA, D21 SCL	2	AI23
	J12	Header Pins (1 × 2) Connectors	5V, GND (IMU)	2	AI25
	J13	Header Pins (1 × 2) Connectors	5V, GND (BTS LEFT LOGIC)	2	P10
	J14	Header Pins (1 × 2) Connectors	5V, GND (BTS RIGHT LOGIC)	2	P17
	R1	Resistor Passives › Resistors	470Ω	2	AJ5
	R2	Resistor Passives › Resistors	10kΩ D30 EN pull-down	2	M7
	R3	Resistor Passives › Resistors	10kΩ D31 EN pull-down	2	M14

03 Shopping list

19 different parts, 25 parts in all. Tick them off as you get them.

	QTY		PART	VALUE	USED AS
☐	1 ×		Resistor Passives › Resistors	10kΩ D30 EN pull-down	R2
☐	1 ×		Resistor Passives › Resistors	10kΩ D31 EN pull-down	R3
☐	1 ×		Resistor Passives › Resistors	470Ω	R1
☐	7 ×		Ceramic Capacitor Passives › Capacitors	100nF X7R	C3 C4 C5 C6 C7 C8 C9
☐	1 ×		Electrolytic Capacitor Passives › Capacitors	220µF 25V (- at from, + at to)	C2
☐	1 ×		Female Header (2.54 mm) (2 × 18) Connectors	MEGA: 1 D32, 2 D33, 3 D2, 4 D22, 5 D3, 6 D23, 7 D19, 8 D25, 9 D18, 10 D24, 11 D5, 12 D6, 13 D30, 14 D9, 15 D10, 16 D20, 17 D21, 18 5V, 19 D34, 20 GND, 21 D31; 22-36 NC	J1
☐	1 ×		Header Pins (1 × 2) Connectors	5V, GND (BTS LEFT LOGIC)	J13
☐	1 ×		Header Pins (1 × 2) Connectors	5V, GND (BTS RIGHT LOGIC)	J14
☐	1 ×		Header Pins (1 × 2) Connectors	5V, GND (IMU)	J12
☐	1 ×		Header Pins (1 × 2) Connectors	D20 SDA, D21 SCL	J11
☐	1 ×		Header Pins (1 × 4) Connectors	D2, D22, 5V, GND	J5
☐	1 ×		Header Pins (1 × 4) Connectors	D3, D23, 5V, GND	J6
☐	1 ×		Header Pins (1 × 4) Connectors	D5, D6, D30, D30	J9
☐	1 ×		Header Pins (1 × 4) Connectors	D9, D10, D31, D31	J10
☐	1 ×		Header Pins (1 × 4) Connectors	D18, D24, 5V, GND	J7
☐	1 ×		Header Pins (1 × 4) Connectors	D19, D25, 5V, GND	J8
☐	1 ×		Header Pins (3 × 2) Connectors	D32, GND, D33, GND, LED_D34_AFTER_R1, GND	J2
☐	1 ×		Header Pins (5 × 1) Connectors	GND BUS FROM MEGA	J4

	QTY	PART	VALUE	USED AS
☐	1 ×	Header Pins (6 × 1) Connectors	5V BUS FROM MEGA ONLY	J3

04 Nets

Each group of pins the board joins. On the finished board, a meter should beep between any two pins of one net, and between no two nets.

NET	PINS IT JOINS
Net (C2.-)	C2.- C3.2 C4.2 C5.2 C6.2 C7.2 C8.2 C9.2 J1.20 J2.2 J2.4 J2.6 J4.1 J4.2 J4.3 J4.4 J4.5 J5.4 J6.4 J7.4 J8.4 J12.2 J13.2 J14.2 R2.2 R3.2
Net (C2.+)	C2.+ C3.1 C4.1 C5.1 C6.1 C7.1 C8.1 C9.1 J1.18 J3.1 J3.2 J3.3 J3.4 J3.5 J3.6 J5.3 J6.3 J7.3 J8.3 J12.1 J13.1 J14.1
Net (J1.1)	J1.1 J2.1
Net (J1.2)	J1.2 J2.3
Net (J1.3)	J1.3 J5.1
Net (J1.4)	J1.4 J5.2
Net (J1.5)	J1.5 J6.1
Net (J1.6)	J1.6 J6.2
Net (J1.7)	J1.7 J8.1
Net (J1.8)	J1.8 J8.2
Net (J1.9)	J1.9 J7.1
Net (J1.10)	J1.10 J7.2
Net (J1.11)	J1.11 J9.1
Net (J1.12)	J1.12 J9.2
Net (J1.13)	J1.13 J9.3 J9.4 R2.1
Net (J1.14)	J1.14 J10.1
Net (J1.15)	J1.15 J10.2
Net (J1.16)	J1.16 J11.1
Net (J1.17)	J1.17 J11.2
Net (J1.19)	J1.19 R1.2
Net (J1.21)	J1.21 J10.3 J10.4 R3.1
Net (J2.5)	J2.5 R1.1

05 Checks

What the editor's checks find on this board.

15 pins not connected

Fine for pins left unused on purpose; otherwise they still need wiring.

J1.22 J1.23 J1.24 J1.25 J1.26 J1.27 J1.28 J1.29 J1.30 J1.31 J1.32 J1.33 J1.34 J1.35 J1.36